Sleep Quality Prediction

Using Machine Learning to Decode Lifestyle Impacts

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SENG 352 Data Analysis Project

Project Motivation

Understanding the science of a good night's rest.

Problem Statement

Sleep quality is a cornerstone of mental and physical health.

Our project addresses the following:

- Quantifying the impact of **Stress Levels** on sleep duration.
- Identifying key lifestyle triggers for poor sleep quality.
- Predicting "Good" vs "Poor" sleep patterns using **ML Classifiers**.
- Assisting individuals in lifestyle optimization through data-driven insights.

Dataset Overview

Feature Category	Key Variables	Data Type
Sleep Metrics	Duration, Quality (1-10), Disorders	Numeric / Categorical
Lifestyle	Stress Level, Physical Activity, Daily Steps	Numeric
Health	BMI, Heart Rate, Blood Pressure	Numeric / Mixed
Demographics	Age, Gender, Occupation	Categorical

Data Preprocessing

Cleaning

Filled missing values in 'Sleep Disorder' with 'None'. Handled duplicates and outliers.

Parsing

Split 'Blood Pressure' string into separate Systolic and Diastolic numeric columns.

Encoding

Converted categorical features (Gender, BMI) into numerical formats using Label Encoding.

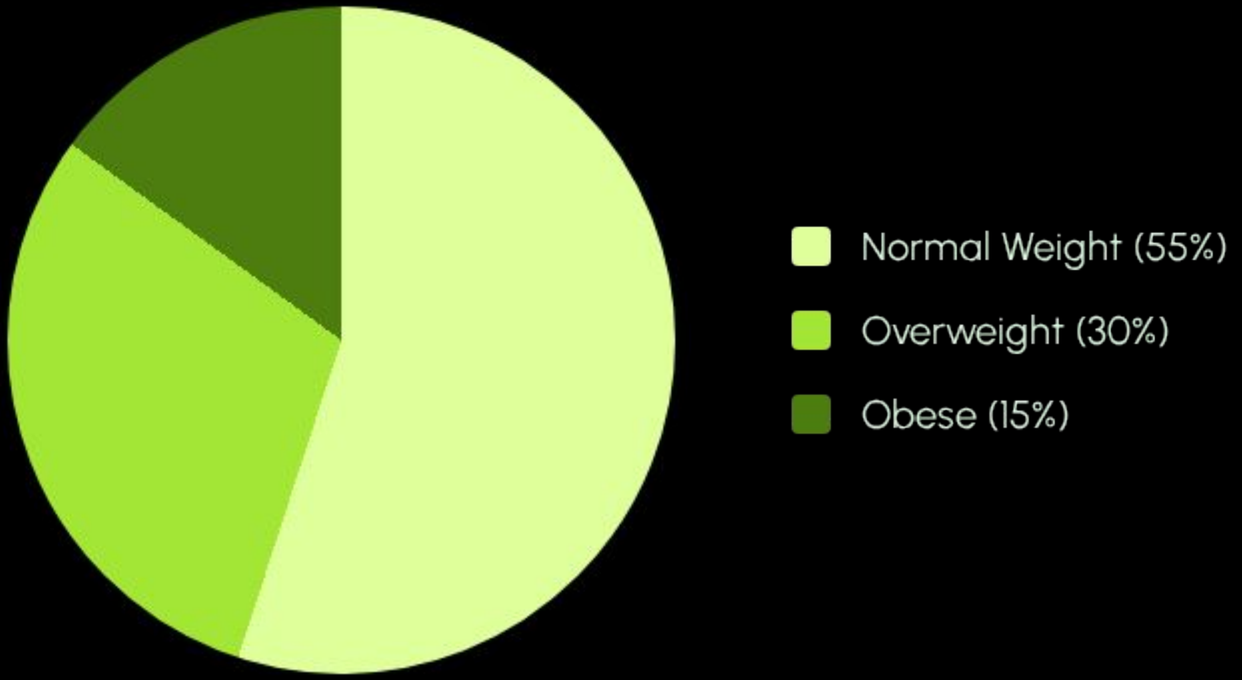
Exploratory Data Analysis

Revealing hidden patterns in health and lifestyle metrics.

The Stress-Sleep Connection

Observation: A dramatic "staircase" effect shows sleep quality degrades as stress increases.

BMI Category Distribution



Modeling Strategy

Baseline Models

We started with **Logistic Regression** to establish a performance floor. This model handles linear relationships effectively.

Advanced Classifiers

Applied **Random Forest** and **Decision Trees** to capture non-linear interactions between variables.

Model Performance

The models achieved near-perfect accuracy due to strong feature correlations (Stress vs Quality).

Project Roadmap

Kickoff
Proposal Approval

Data
Acquisition & Cleaning

EDA
Analysis & Visualization

ML
Training & Optimization

Final
Presentation & Code

Thank You

Any Questions?

Project Repository:

github.com/beyzaxozdemir/sleep-prediction

Image Sources

https://png.pngtree.com/thumb_back/fw800/background/20250903/pngtree-futuristic-medical-chamber-with-woman-observing-hightech-bed-image_16777611.jpg

Source: pngtree.com